Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train.

1. Introduction

Recurrent models typically factor computation along the symbol positions of the input and output sequences. Aligning the positions to steps in computation time, they generate a sequence of hidden states h_t, as a function of the previous hidden state h_{t-1} and the input for position t. This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths.

Model Architecture & Self-Attention

Section 3: Attention Mechanism and Multi-Head Projections

3.1 Scaled Dot-Product Attention

We compute the attention function on a set of queries simultaneously, packed together into a matrix Q. The keys and values are packed into matrices K and V. We compute the dot products of the query with all keys, divide each by the square root of the key dimension d_k, and apply a softmax function to obtain the weights on the values:

`Attention(Q, K, V) = softmax(Q K^T / sqrt(d_k)) V`

3.2 Multi-Head Attention

Instead of performing a single attention function with d_model-dimensional keys, values and queries, we found it beneficial to linearly project the queries, keys and values h times with different, learned linear projections to d_k, d_k and d_v dimensions, respectively. For the base model, we employ h = 8 parallel attention layers or heads. For each of these, we use d_k = d_v = d_model / h = 64. On each of these projected versions of queries, keys and values we then perform the attention function in parallel.

Experimental Results & Limitations

Section 5: Evaluation, Empirical Benchmarks and Limitations

5.1 Translation Benchmarks Table

Below are the empirical BLEU scores and training costs across translation benchmarks compared to prior recurrent and convolutional architectures:

Model	WMT14 En-De	WMT14 En-Fr	Training Time
ByteNet	23.75	-	-
ConvS2S	25.16	40.46	9.6 days (8 P100)
Transformer (base)	27.30	38.10	12 hours (8 P100)
Transformer (big)	28.40	41.80	3.5 days (8 P100)

6. Reported Limitations

A key reported limitation is that standard self-attention has a computational complexity and memory footprint of O(n^2) with respect to the sequence length n. This quadratic complexity limits direct application of standard transformers to extremely long context windows without specialized sparse or linear approximation techniques.